

Canonical processes

Concrete substochastic tensors $T^{(x)}$ + state-diagram automata for the small HMMs used in computational mechanics and in Shai et al. 2024. Each entry presents (i) the parametrised, symbol-labelled transition tensors with row-sum check, (ii) the corresponding HMM automaton as a `stateDiagram-v2` (Mermaid) graph rendering states as nodes and transitions as `symbol / probability`-labelled edges, and (iii) the stationary distribution and reachable-belief structure.

Notation per [notation.md](#): row vectors act on the right, $T_{ij}^{(x)} = \Pr(X_t = x, S_{t+1} = s_j \mid S_t = s_i)$. Substochasticity: $\sum_x \sum_j T_{ij}^{(x)} = 1$ per row.

Mermaid edge labels use the form `symbol / probability`. Multiple emissions on one edge are comma-separated. Self-loops are rendered as nodes pointing to themselves.

1. Binary IID

Description. Single-state HMM emitting independent Bernoulli(p) samples. The trivial baseline against which every non-Markov-order-1 process is compared. Process is its own ε -machine.

States. $\mathcal{S} = \{S\}$.

Bias parameter $p \in (0, 1) = \Pr(\text{emit } 1)$.

$$T^{(0)} = [1 - p], \quad T^{(1)} = [p].$$

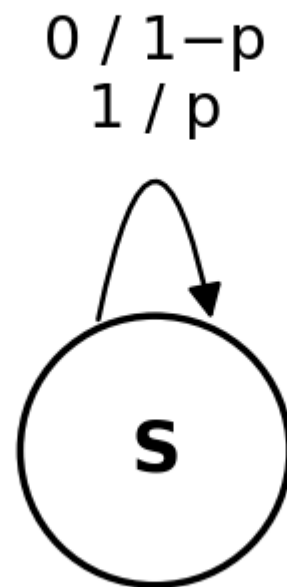
Row-sum check. State S : $(1 - p) + p = 1 \checkmark$.

Stationary distribution. $\pi = (1)$ trivially.

Reachable mixed states. $\mathcal{B}_\infty = \{(1)\}$ — a single point. The ε -machine is one-state and the MSP is degenerate.

Information rates. $h_\mu = -p \log p - (1 - p) \log(1 - p) = H(p)$ (binary entropy); $C_\mu = 0$; $E = 0$.

Automaton.



2. Golden-mean process

Description. Binary process with forbidden substring **11**. Canonical sofic shift; a Markov chain on observable states $\{0, 1\}$ with the constraint that **1** cannot follow **1**. (Crutchfield & Young 1989; Shalizi & Crutchfield 2001.)

States. $\mathcal{S} = \{A, B\}$. A = "last symbol was **0** (or start)"; B = "last symbol was **1**".

Bias parameter $p \in (0, 1) = \Pr(\text{emit } 1 \mid A)$.

$$T^{(0)} = \begin{bmatrix} 1-p & 0 \\ 1 & 0 \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} 0 & p \\ 0 & 0 \end{bmatrix}.$$

Row-sum check (with $p = 0.5$): both states sum to **1** ✓.

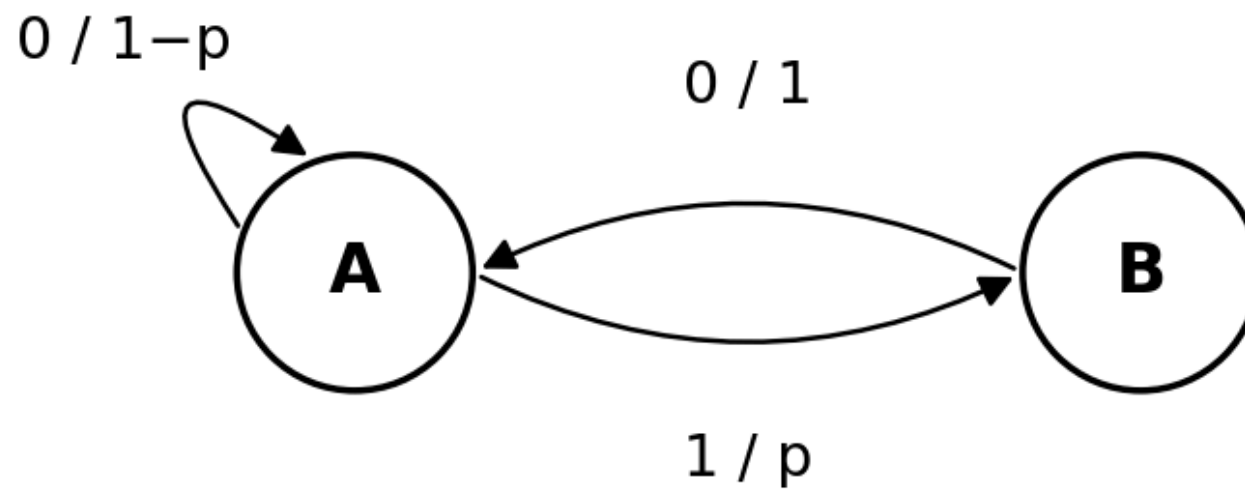
Stationary distribution. $T = T^{(0)} + T^{(1)} = \begin{bmatrix} 1-p & p \\ 1 & 0 \end{bmatrix}$. Solving $\pi T = \pi$:

$$\pi = \left(\frac{1}{1+p}, \frac{p}{1+p} \right).$$

For $p = 0.5$: $\pi = (2/3, 1/3)$.

Reachable mixed states. Two points after one observation; MSP is unifilar 2-state.

Automaton.



3. Even process

Description. Binary process; **1**s come in even-length blocks. Canonical non-Markov-order-1 sofic example: not Markov of any finite order on observed symbols, but a 2-state HMM. (Shalizi & Crutchfield 2001.)

States. $\mathcal{S} = \{A, B\}$. A = "just saw an even number of **1**s since the last **0** (or start)"; B = "just saw a **1**, parity incomplete".

Bias parameter $p \in (0, 1) = \Pr(\text{emit } 1 \mid A)$.

$$T^{(0)} = \begin{bmatrix} 1-p & 0 \\ 0 & 0 \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} 0 & p \\ 1 & 0 \end{bmatrix}.$$

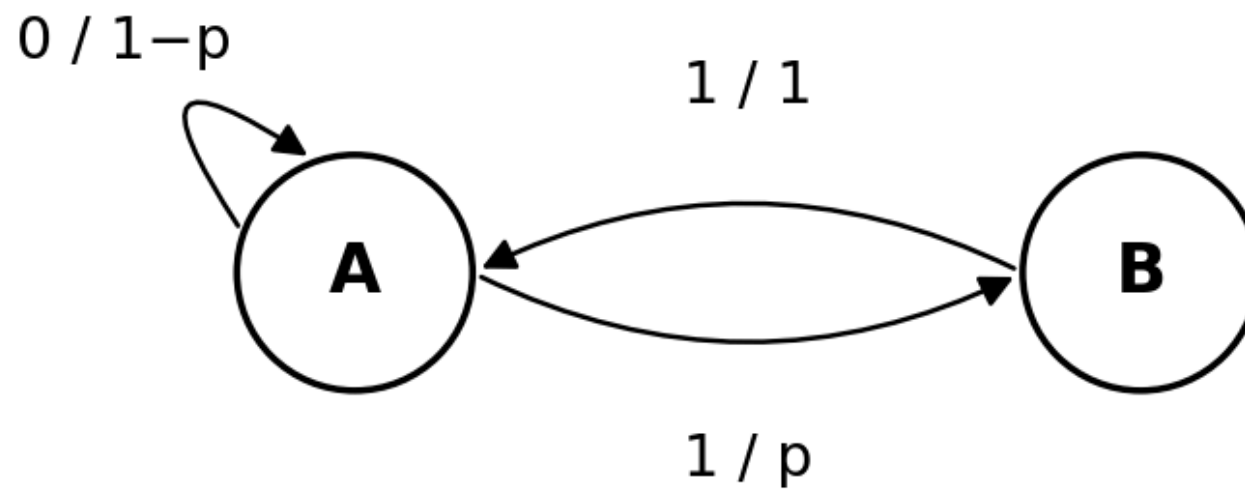
Row-sum check (with $p = 0.5$):

- State A : $(1-p) + 0 + 0 + p = 1 \checkmark$
- State B : $0 + 0 + 1 + 0 = 1 \checkmark$

Stationary distribution. Same Markov chain on states as golden mean: $\pi = \left(\frac{1}{1+p}, \frac{p}{1+p} \right)$. The two processes share the underlying T but assign symbols differently — the *emission* labels distinguish them, not the dynamics on $\mathcal{S}$.

Reachable mixed states. Finitely many; MSP is a 2-state HMM.

Automaton.



The asymmetry with golden-mean: state *B* emits **1** deterministically (completing the even block) rather than **0**.

4. Simple Nonunifilar Source (SNS)

Description. The canonical *non-unifilar* HMM: from one state, the same emission can lead to two different next states. The smallest non-trivial example showing that non-unifilar HMMs typically have an *infinite* mixed-state set $\mathcal{B}_\infty$ even though the HMM itself has only two states. (Crutchfield 1994; Crutchfield & Feldman 2003; Marzen & Crutchfield 2017.)

States. $\mathcal{S} = \{A, B\}$.

Bias parameter $p \in (0, 1)$ — the "stay" probability shared by both states. Each state has self-loop probability p and transition-out probability $1 - p$.

$$T^{(0)} = \begin{bmatrix} p & 1-p \\ 0 & p \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} 0 & 0 \\ 1-p & 0 \end{bmatrix}.$$

Row-sum check.

- State A : $p + (1-p) + 0 + 0 = 1$ ✓
- State B : $0 + p + (1-p) + 0 = 1$ ✓

Non-unifilarity. From state A , emission 0 leads to *either* A (prob p) or B (prob $1-p$): $T_{A,A}^{(0)} > 0$ and $T_{A,B}^{(0)} > 0$ — the (state, symbol) pair does not uniquely determine the next state. State B is unifilar: emitting 0 uniquely keeps it in B , emitting 1 uniquely sends it to A . The non-unifilarity is *only* at A .

Asymmetry of the alphabet. Symbol 1 is "diagnostic" — observing it definitely came from B ending up in A . Symbol 0 is "ambiguous" — it could have come from A (staying or moving to B) or from B (staying). This is what makes the belief state evolve non-trivially under runs of 0 s.

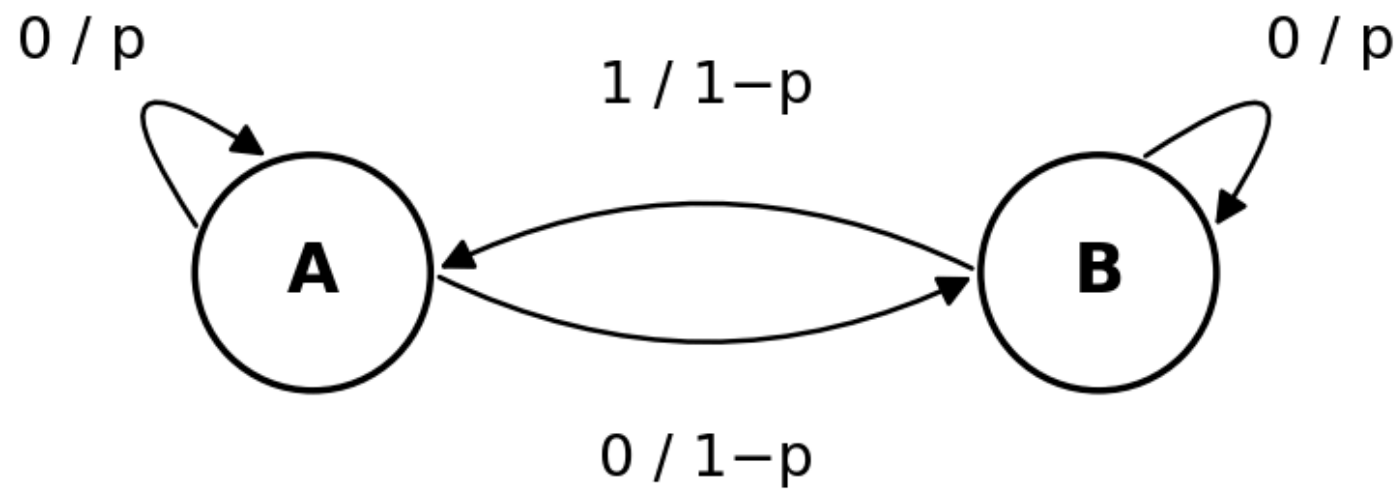
Stationary distribution. $T = T^{(0)} + T^{(1)} = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$. The full transition is *doubly stochastic* (every column also sums to 1 by symmetry), so the stationary is uniform:

$$\pi = \left(\frac{1}{2}, \frac{1}{2}\right) \quad \forall p \in (0, 1).$$

Reachable mixed states. Infinite — $\mathcal{B}_\infty$ is a Cantor-like subset of Δ^1 (the 1-simplex $[0, 1]$). Each **0**-observation feeds the belief through a non-trivial Bayesian update because the emission is ambiguous; each **1**-observation deterministically resets the belief to "just transitioned $B \rightarrow A$, now in A ". So $\mathcal{B}_\infty$ contains a discrete "post-1" attractor point at $\eta = (1, 0)$ plus a fractal continuum of "post-run-of-0s" beliefs. *This is the smallest example demonstrating that a finite HMM can have an infinite ε -machine.*

Information rates. $C_\mu < \infty$ on the HMM presentation, but the ε -machine has $C_\mu^{(\varepsilon)} = \infty$. Marzen & Crutchfield 2017 give the closed-form fractal dimension of $\mathcal{B}_\infty$.

Automaton.



The two outgoing edges from **A** both bear the same emission symbol **0** — visually, this *is* the non-unifilarity.

5. Fractal 2-state

Description. A 3-parameter family of 2-state non-unifilar HMMs. Both states are non-unifilar (one on each symbol), so the belief evolves through *two* contracting non-affine maps on Δ^1 , producing a Cantor-like attractor. Generalises the SNS by giving each state its own non-unifilar branch on the "wrong" emission, with an independent emission bias.

States. $\mathcal{S} = \{A, B\}$.

Parameters $(b, p, q) \in (0, 1)^3$. The kernel is built from emissions and conditional next-states:

- **Emission bias** $b = \Pr(0 \mid A) = \Pr(1 \mid B)$ — each state prefers its "matching" symbol with probability b .
- **Leaving A on 1:** $p = \Pr(B \mid 1, A)$, so $1 - p = \Pr(A \mid 1, A)$.
- **Leaving B on 0:** $q = \Pr(A \mid 0, B)$, so $1 - q = \Pr(B \mid 0, B)$.
- The two "matching" emissions are deterministic: $\Pr(A \mid 0, A) = 1$ and $\Pr(B \mid 1, B) = 1$.

Both p and q now represent the probability of *leaving* the current state on the "wrong" emission — they play structurally identical roles for A and B .

The substochastic kernel is $T_{r \rightarrow r'}^{(x)} = p(x \mid r) p(r' \mid x, r)$:

$$T^{(0)} = \begin{bmatrix} b & 0 \\ (1-b)q & (1-b)(1-q) \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} (1-b)(1-p) & (1-b)p \\ 0 & b \end{bmatrix}.$$

Row-sum check.

- State A : $b + 0 + (1-b)(1-p) + (1-b)p = b + (1-b) = 1 \checkmark$
- State B : $(1-b)q + (1-b)(1-q) + 0 + b = (1-b) + b = 1 \checkmark$

Non-unifilarity. *Both* states are non-unifilar — each on the "wrong" emission:

- From A emitting 1: next state is A (prob $1 - p$) or B (prob p).
- From B emitting 0: next state is A (prob q) or B (prob $1 - q$).

The "matching" emissions are unifilar: A on 0 stays in A deterministically, B on 1 stays in B deterministically. This is the structural difference from §4 SNS: here, *both* states have a non-unifilar branch, doubling the contraction-map count and producing a richer fractal attractor.

Stationary distribution. Full transition $T = T^{(0)} + T^{(1)}$ has

$$T = \begin{bmatrix} b + (1-b)(1-p) & (1-b)p \\ (1-b)q & b + (1-b)(1-q) \end{bmatrix}.$$

Solving $\pi T = \pi$:

$$\pi = \left(\frac{q}{p+q}, \frac{p}{p+q} \right).$$

When $p = q$ (symmetric branching), $\pi = (1/2, 1/2)$ for any p — the symmetric parameterisation gives uniform stationary regardless of branching strength.

Reachable mixed states. $\mathcal{B}_\infty$ is generically a Cantor-like *fractal* subset of Δ^1 . The belief-update IFS has two non-affine contractions; parameterising $\eta = (\alpha, 1 - \alpha)$,

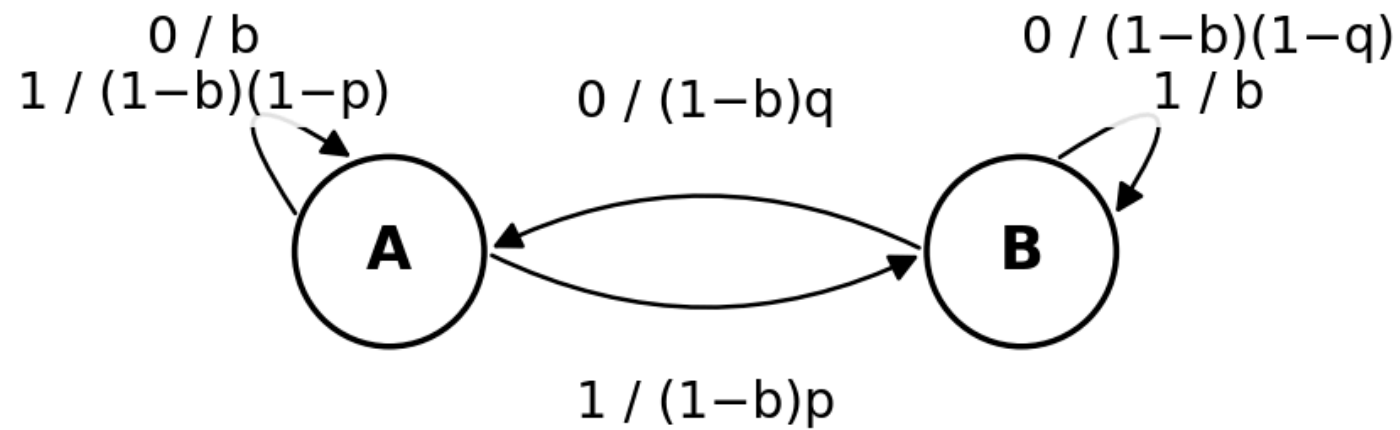
$$f_0(\alpha) = \frac{\alpha b + (1 - \alpha)(1 - b)q}{\alpha b + (1 - \alpha)(1 - b)}, \quad f_1(\alpha) = \frac{\alpha(1 - b)(1 - p)}{\alpha(1 - b) + (1 - \alpha)b}.$$

For generic (b, p, q) with both branches genuinely stochastic (i.e. $p, q \in (0, 1)$), the attractor has Hausdorff dimension strictly less than 1.

Boundary cases (collapse to unifilar):

- $p \rightarrow 0$: from A on 1 , always stay in A (deterministic).
- $p \rightarrow 1$: from A on 1 , always go to B (deterministic).
- $q \rightarrow 0$: from B on 0 , always stay in B (deterministic).
- $q \rightarrow 1$: from B on 0 , always go to A (deterministic).
- Any of these eliminate one contraction; if both p, q are at their boundaries, $\mathcal{B}_\infty$ becomes finite.

The "fractal" regime is when p, q are properly interior to $(0, 1)$.

Automaton.

Both states have *two-emission self-loops* (one for each symbol), and there is exactly one outgoing inter-state edge from each state — labelled with the "wrong" emission for that state and bearing the only non-deterministic branching.

6. Zero-one-random (Z1R)

Description. Period-3 deterministic skeleton $0 \rightarrow 1 \rightarrow R \rightarrow 0 \rightarrow \dots$ where the third position is a fair coin. (Shalizi & Crutchfield 2001.)

States. $\mathcal{S} = \{S_0, S_1, S_R\}$, ordered.

$$T^{(0)} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0.5 & 0 & 0 \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0.5 & 0 & 0 \end{bmatrix}.$$

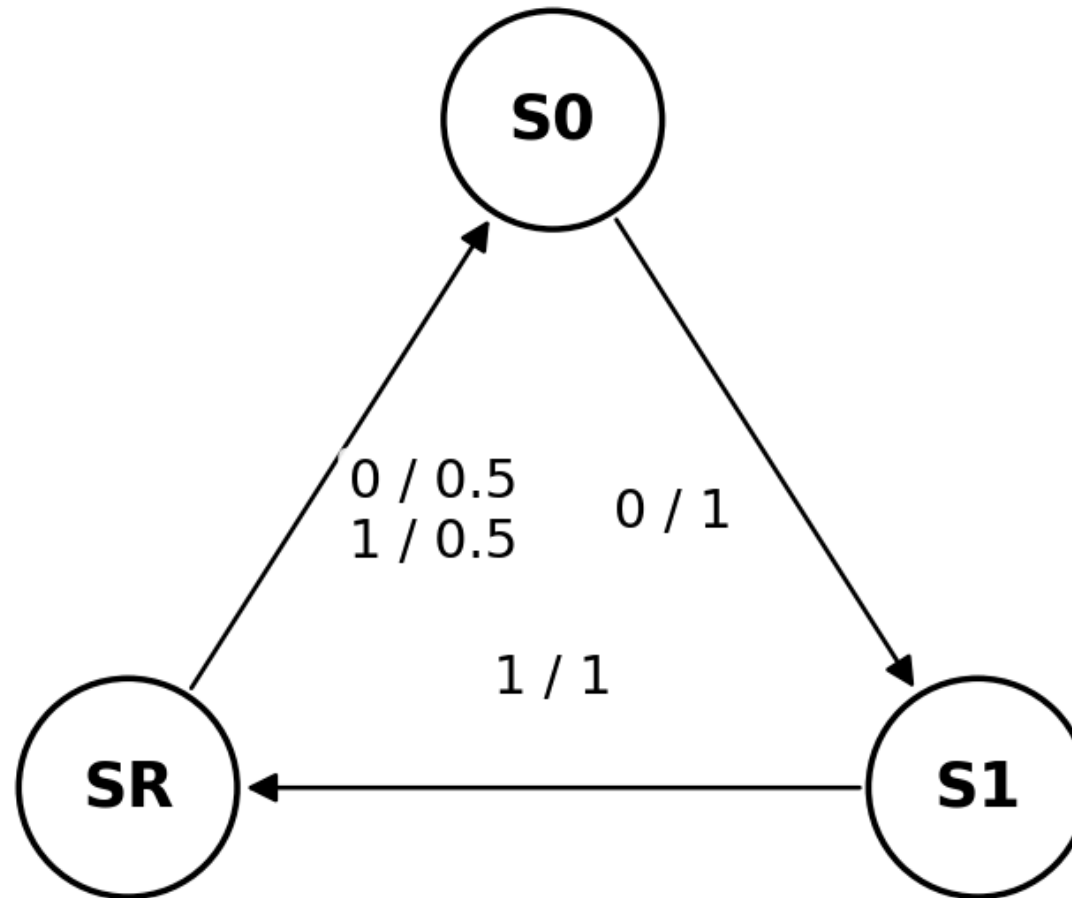
Row-sum check.

- S_0 : $1 + 0 = 1 \checkmark$
- S_1 : $0 + 1 = 1 \checkmark$
- S_R : $0.5 + 0.5 = 1 \checkmark$

Stationary distribution. By the period-3 cycle, $\pi = (1/3, 1/3, 1/3)$.

Reachable mixed states. 3 vertex-like points after synchronisation.

Automaton.



[check: row ordering convention — $T_{S_R, S_0}^{(x)} = 0.5$ for both $x = 0$ and $x = 1$. Verify against Shalizi & Crutchfield 2001 §6 before using in code.]

7. Mess3

Description. 3-state, 3-symbol HMM whose MSP has fractal structure on the 2-simplex; the canonical Shai et al. (2024) Fig. 5 demonstration of "linearly encoded fractal belief states in a transformer residual stream".

States. $\mathcal{S} = \{S_0, S_1, S_2\}$ (labels per Shai et al. §A.1).

Parameterisation (x, α) , with derived $b = (1 - \alpha)/2$ and $y = 1 - 2x$:

$$T^{(A)} = \begin{bmatrix} \alpha y & bx & bx \\ \alpha x & by & bx \\ \alpha x & bx & by \end{bmatrix},$$

$$T^{(B)} = \begin{bmatrix} by & \alpha x & bx \\ bx & \alpha y & bx \\ bx & \alpha x & by \end{bmatrix}, \quad T^{(C)} = \begin{bmatrix} by & bx & \alpha x \\ bx & by & \alpha x \\ bx & bx & \alpha y \end{bmatrix}.$$

The cyclic symmetry: $T^{(B)}$ is obtained from $T^{(A)}$ by cycling (state, symbol) labels $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$, and $T^{(C)}$ by cycling once more.

Paper instance. Shai et al. (2024) §A.1 use $(x, \alpha) = (0.05, 0.85)$, giving $\alpha y = 0.765$, $\alpha x = 0.0425$, $by = 0.0675$, $bx = 0.00375$. Reference repo `adamimos/epsilon-transformers` defaults to $(x, \alpha) = (0.15, 0.6)$, which yields a *different* fractal — verified 2026-04-21. Our [src/processes.py](#) `mess3()` defaults to $(0.05, 0.85)$ to match the paper.

Numerical instance at paper params:

$$T^{(A)} = \begin{bmatrix} 0.765 & 0.00375 & 0.00375 \\ 0.0425 & 0.0675 & 0.00375 \\ 0.0425 & 0.00375 & 0.0675 \end{bmatrix},$$

$$T^{(B)} = \begin{bmatrix} 0.0675 & 0.0425 & 0.00375 \\ 0.00375 & 0.765 & 0.00375 \\ 0.00375 & 0.0425 & 0.0675 \end{bmatrix}, \quad T^{(C)} = \begin{bmatrix} 0.0675 & 0.00375 & 0.0425 \\ 0.00375 & 0.0675 & 0.0425 \\ 0.00375 & 0.00375 & 0.765 \end{bmatrix}.$$

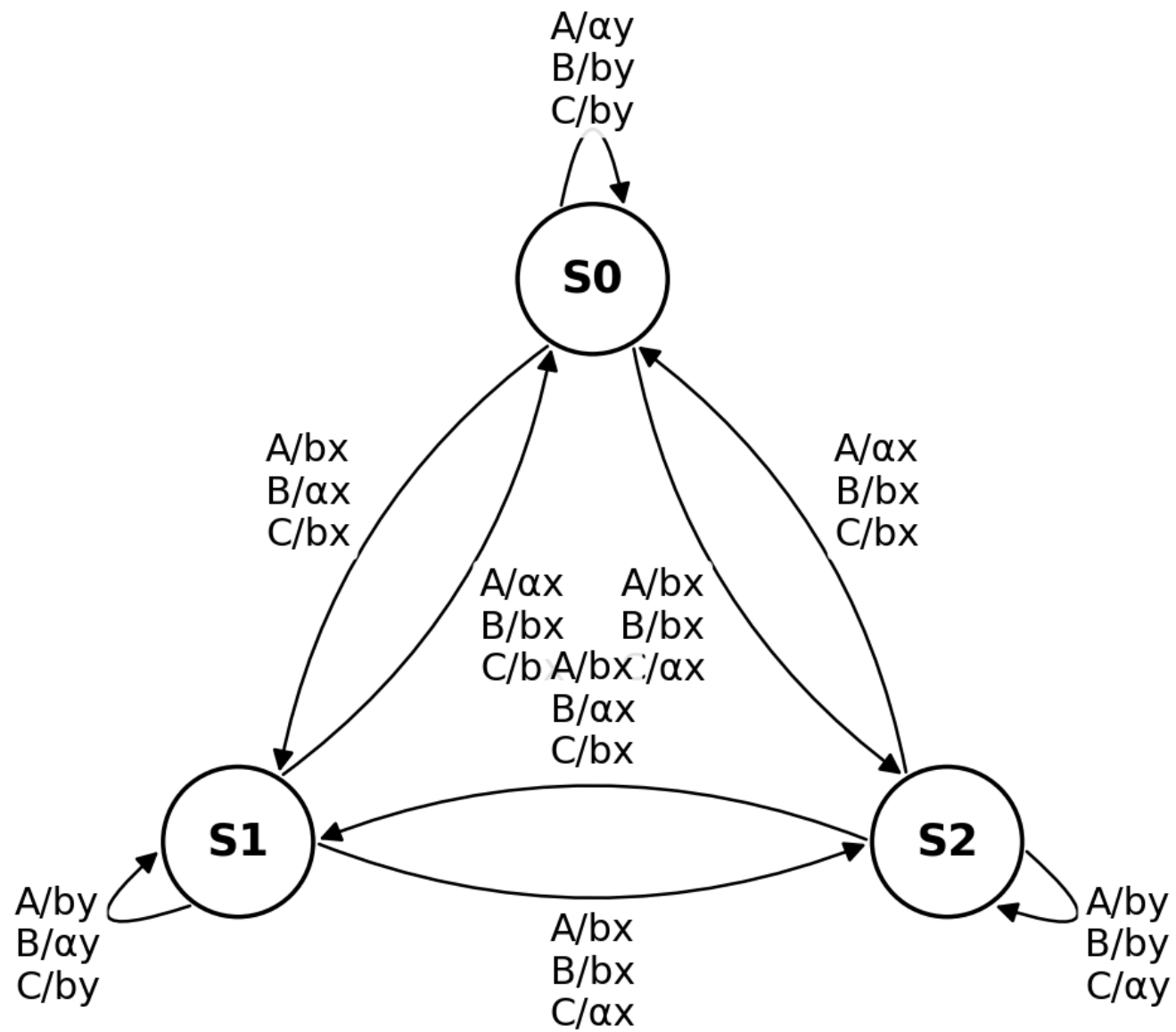
Row-sum check. Each state's row across all three matrices sums to 1:

- State S_0 : $0.7725 + 0.11375 + 0.11375 = 1.00000 \checkmark$
- State S_1 : by cyclic symmetry = $1.00000 \checkmark$
- State S_2 : by cyclic symmetry = $1.00000 \checkmark$

Stationary distribution. By cyclic symmetry $\pi = (1/3, 1/3, 1/3)$.

Reachable mixed states. $|\mathcal{B}_\infty| = \infty$. The attractor is a fractal subset of Δ^2 resembling a deformed Sierpinski gasket. Closed-form Hausdorff dimension not recorded in Shai et al.; conjecturally given by Marzen & Crutchfield 2017's general dimension–spectrum scaling, not yet evaluated for Mess3 specifically.

Automaton (parameterised; full kernel above).



The dominant transitions are the αy self-loops (state S_i emits the matching symbol X_i and stays). The αx transitions are the "synchronising" moves (any state, emit symbol X_i , move toward S_i). The bx transitions are the small "random walk" between non-matching state-symbol pairs.

8. RRXOR (Random-Random-XOR)

Description. 5-state binary HMM. Two fair random bits R_1, R_2 followed by their XOR, then repeat. Shai et al. (2024) Fig. 7. Designed to exhibit *crypticity* $\chi = C_\mu - E > 0$: belief states share their next-token marginals but differ on full-future joints, so the structure is hidden from any 1-step predictor.

States. $\mathcal{S} = \{0, 1, 2, 3, 4\}$, zero-indexed. State 0 is pre- R_1 ; states 1, 2 are post- R_1 ; states 3, 4 emit the XOR result deterministically.

Transition tensors (Shai et al. §A.1):

$$T^{(0)} = \begin{bmatrix} 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0.5 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Row-sum check.

State	Row	Row	Sum
0	0.5	0.5	1.0 ✓
1	0.5	0.5	1.0 ✓
2	0.5	0.5	1.0 ✓

State	Row($T^{(0)}$)	Row($T^{(1)}$)	Sum
3	0	1	1.0 ✓
4	1	0	1.0 ✓

XOR sanity-check (traversing the state graph):

(R_1, R_2)	path	step-3 emission
(0, 0)	$0 \xrightarrow{0} 1 \xrightarrow{0} 4 \xrightarrow{0} 0$	0 ✓
(0, 1)	$0 \xrightarrow{0} 1 \xrightarrow{1} 3 \xrightarrow{1} 0$	1 ✓
(1, 0)	$0 \xrightarrow{1} 2 \xrightarrow{0} 3 \xrightarrow{1} 0$	1 ✓
(1, 1)	$0 \xrightarrow{1} 2 \xrightarrow{1} 4 \xrightarrow{0} 0$	0 ✓

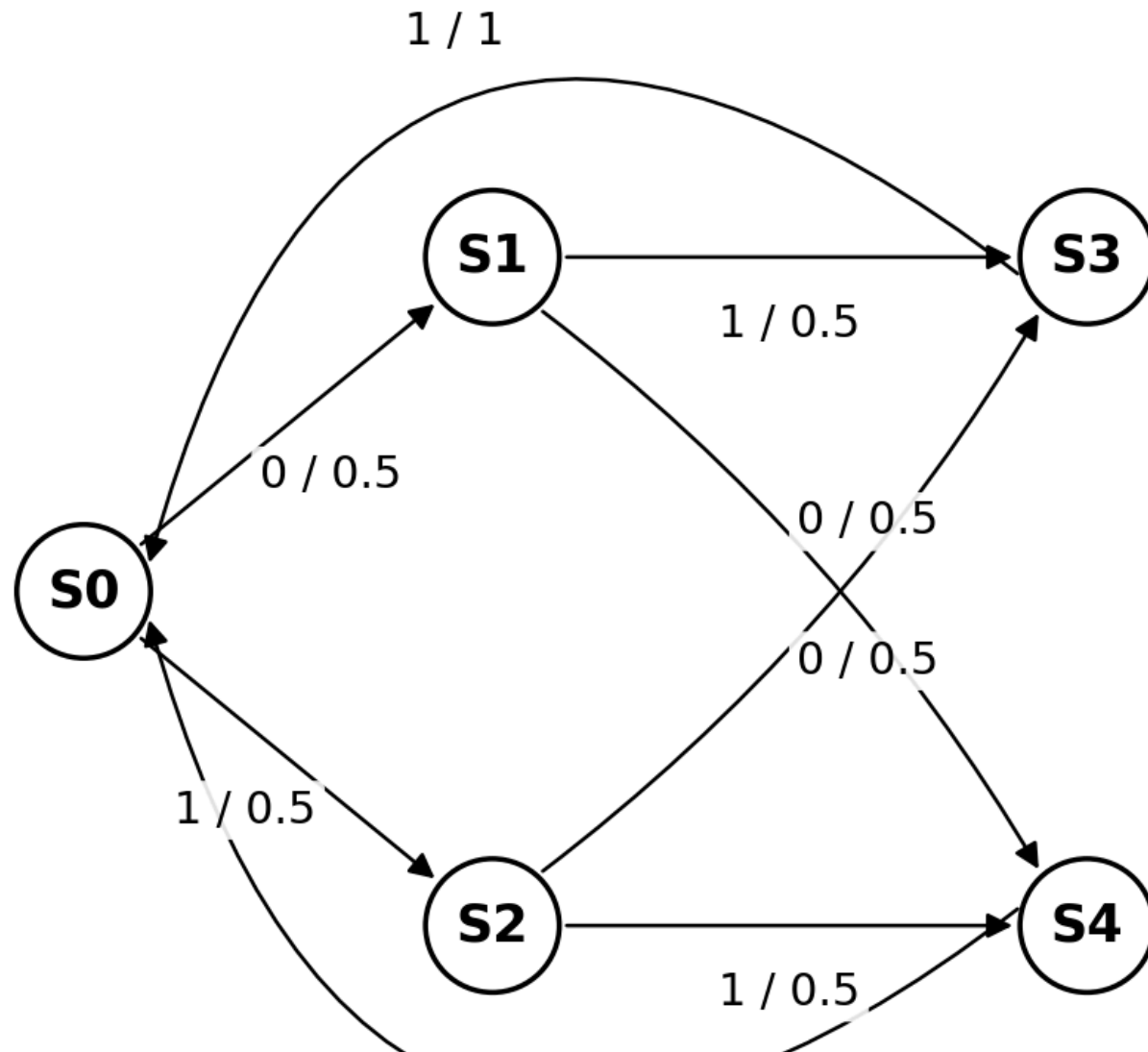
Stationary distribution. By the period-3 sample paths,

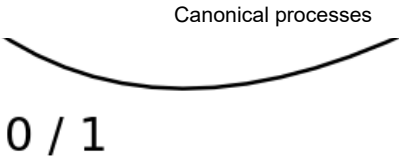
$$\pi = \left(\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right).$$

Check: πT at state 0 gets $1 \cdot \frac{1}{6}$ (from state 3) + $1 \cdot \frac{1}{6}$ (from state 4) = $\frac{1}{3}$ ✓. Remaining by symmetry.

Reachable mixed states. $|\mathcal{B}_\infty| = 36$ (Shai et al. §3.2). Multiple beliefs share next-token distribution but differ on full-future joints — the crypticity phenomenon RR XOR is designed to exhibit.

Automaton.





[check: reproduce the enumeration $|\mathcal{B}_\infty| = 36$ in code and confirm the paper's partition into next-token-equivalence classes — this is the unit test for the whole probing pipeline.]

9. Summary table

Process		Alphabet	Unifilar?		MSP geometry	Purpose
Binary IID	1	$\{0, 1\}$	Yes (trivially)	1	one point	baseline; $C_\mu = 0$
Golden-mean	2	$\{0, 1\}$	Yes	2	2 points	sofic forbidden-word
Even	2	$\{0, 1\}$	Yes	2	2 points	minimal non-Markov sofic
SNS	2	$\{0, 1\}$	No (A)	∞	Cantor-like on Δ^1	smallest non-unifilar example (1 contraction)
Fractal 2-state	2	$\{0, 1\}$	No (A, B)	∞	fractal on Δ^1	3-param non-unifilar (2 contractions)

Process	$ \mathcal{S} $	Alphabet	Unifilar?	$ \mathcal{B}_\infty $	MSP geometry	Purpose
Z1R	3	$\{0, 1\}$	Yes	3	3 vertices	period + randomness
Mess3	3	$\{A, B, C\}$	Yes	∞	fractal on Δ^2	Shai et al. primary demo
RRXOR	5	$\{0, 1\}$	Yes	36	36 points in Δ^4	crypticity demo

The unifilarity column matters operationally: unifilar = (state, symbol) determines next state, so the ε -machine is finite-state on the original $\mathcal{S}$. SNS is the one entry where this fails — and consequently, the only entry where $|\mathcal{B}_\infty| = \infty$ for a *finite* HMM (Mess3 has $|\mathcal{B}_\infty| = \infty$ because the MSP is fractal even though the HMM itself is unifilar).

10. Implementations

Process	Code	Notes
Mess3	src/processes.py <code>mess3(x, a)</code>	cross-checked against simplexity 2026-04-28
RRXOR	src/processes.py <code>rrxor(pR1, pR2)</code>	$ \mathcal{B}_\infty = 36$ unit test pending
Even, Golden-mean, Z1R, Binary IID, SNS, Fractal 2-state	not yet	trivial to add as parameter-only HMM($T=\dots$) builders

11. Notation conventions for the diagrams

- Edge label form: `symbol / probability` . Multi-symbol edges (where multiple emissions share the same source/target pair) stack the labels on separate lines.
- Substochasticity: row sums (over symbols $\times$ next-states) = **1** per state. The diagrams make this visible — every state's outgoing edges sum to **1** when probabilities are summed.

- Where a state has a self-loop (Binary IID, Mess3, Even, Golden-mean, SNS, Fractal 2-state), the loop is drawn as the state pointing to itself.
- For Mess3, edge labels are *symbolic* in (x, α) rather than numerical, so the same diagram covers the paper instance and the reference-repo default. The accompanying caption gives the numerical mapping at $(0.05, 0.85)$. Fractal 2-state is similarly parameterised in (b, p, q) .

Source. All seven figures are rendered from a single deterministic Python script at [figures/render_automata.py](#) using matplotlib + FancyArrowPatch (no graphviz / Mermaid dependency). To regenerate after any tensor change:

```
~/pyenv/versions/3.11.9/bin/python3 handbook/figures/render_automata.py
```